

CHAPTER

8

INTRODUCTION TO LISSAJOUS ORBIT ACQUISITION AND INTERPRETATION

8.0 INTRODUCTION

In recent years, it has become common practice to install dual non-contacting eddy current probes on critical high speed turbomachinery to avoid false shutdowns in the event that one of the probes should fail. The pickups are mounted radially in each bearing with the axis of the pickups separated by 90°. Some are mounted in the true horizontal and vertical directions; others are mounted on an axis 45° from the horizontal at the "10:30" and "1:30" clock positions. Since these non-contacting pickups are already positioned in the bearings, they can be hooked into an oscilloscope to provide Lissajous orbit patterns of the shaft's motion in the bearing.

This chapter will deal with the proper setup of transducers and instrumentation required to produce Lissajous orbits. Some example Lissajous plots will be given and their use in the diagnosis of common machine problems will be provided. The interpretation of Lissajous plots will be the highlight of this chapter.

Since this is an introduction to Lissajous orbit generation and interpretation, advanced techniques and complex situations will not be featured. However it will provide a solid basis for future discussion.

8.01 WHAT IS A LISSAJOUS ORBIT?

A Lissajous orbit is a plot of the shaft's orbit within the bearing corresponding to the shaft's total motion. This orbit will likewise display the total shaft displacement (mils or microns). Perhaps, it is easiest to understand if one focuses on the center of the end of a shaft. If the motion of that center could be magnified and the rotational speed slowed down, one would be able to see how the shaft center is actually moving. Is it jumping erratically? Is it rotating in circular loops? Is it rotating in an elliptical fashion? What does it mean? The purpose of a Lissajous orbit is to provide a trace of the actual movement of the shaft within the confines of the clearance in a plain bearing.

Lissajous "plots can be used to detect abnormal movements of the shaft in a bearing. Abnormal movements will often be a characteristic of a developing fault".¹

8.02 A TYPICAL SETUP FOR GENERATING LISSAJOUS ORBITS

Figure 1 is a typical setup for generating a Lissajous plot. It features two non-contacting pickups, a signal conditioner, and an oscilloscope (many of today's multi-channel data collectors and spectrum analyzers can likewise generate Lissajous orbits). An alternative setup is shown in the upper portion of Figure 2.

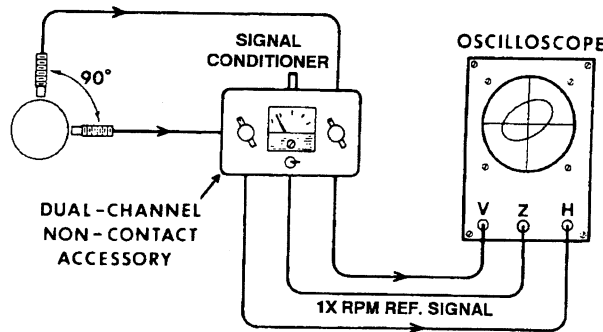


FIGURE 1
A TYPICAL SETUP FOR DISPLAYING LISSAJOUS ORBITS USING
NON-CONTACTING EDDY CURRENT PROBES (Ref. 2)

8.03 SETTING UP THE NON-CONTACT PICKUPS FOR LISSAJOUS ORBITS

It would be ideal if the non-contacting pickups were installed with their axis in the true horizontal and vertical directions at the shaft. In many cases however, they are placed 45° from the horizontal to allow for the split lines in the bearing housings. This will not affect the shape of the observed pattern, but it will cause the display to be rotated 45°. However, compensation for this can be made on the CRT screen on the oscilloscope. How this is accomplished will be mentioned later.

It is important that the sensitivities of the non-contacting transducers be known. It is necessary to check the specifications to be sure. The sensitivity rating is given in millivolts per mil. For example, most non-contact pickups have a sensitivity of 200 millivolts per mil. These input sensitivities are matched by the adjusting the gain sensitivities on the oscilloscope. (Ref. 2)

The reason for this information is to ensure that the shape of the Lissajous orbit accurately reflects the true motion of the shaft. It is important that both the horizontal and vertical scales are set up with identical displacement scaling (.1 mil/division; 1.0 mil/division, etc.). Otherwise, all resultant patterns (orbits) will be distorted. In addition, it is very desirable to ensure accurate amplitude readings from the scope (or analyzer).

8.04 PROVIDING A ONCE-PER-REVOLUTION REFERENCE PULSE

A choice is usually made between installing another non-contacting pickup or an electromagnetic pickup close to the shaft to provide a once per revolution trigger pulse. Usually a key or keyway is used to trigger the reference pickup. If there is no key or keyway, a shallow slot can be made in the shaft in a discrete location to trigger the pickup. Other alternatives are available.

If the oscilloscope is equipped with a "Z" axis (intensity) input, the lead from the reference pickup can be inserted there. This will produce one or more "blank" spots on the Lissajous orbits.

8.05 THE OSCILLOSCOPE - THE CONVENTIONAL CHOICE

For many years, the oscilloscope has been the conventional instrument of choice to produce Lissajous plots. With the advent of multi-channel FFT analyzers, it has become possible to generate Lissajous orbits with only a few problems to overcome. It has been stated that "a well-equipped multichannel FFT analyzer has features which will improve our ability to analyze orbit information, and it can be a more powerful tool than an oscilloscope".² Certainly a single channel FFT analyzer can not be used, but a 2-channel FFT analyzer would be adequate for Lissajous orbits. For instance, this is possible by using the "oscilloscope mode" of such a 2-channel analyzer.

While this may be so, for the scope of the material in this chapter, only the oscilloscope will be featured in this introductory article.

8.06 CONSIDERATION OF PICKUP LOCATION, DIRECTION OF SHAFT MOTION, AND THE POLARITY OF THE POWER SUPPLY

Nearly all oscilloscopes are designed so that an increase in positive (+) voltage of the input signal (or a decrease in minus (-) voltage) will cause the vertical trace to move upwards or the horizontal trace to move to the right. (Ref. 2) Two factors need to be considered.

1. The polarity of the non-contact pickup system.
2. The location of the pickup (above or below; on the right or the left side of the shaft).

If the shaft is moving towards a pickup, it is closing the gap, thus reducing the non-contact pickup voltage; if it moves away from the pickup, it is increasing the gap, thus increasing the voltage (i.e., .200 mv/mil gap applied to the DC voltage). The ideal situation is that in whatever direction the shaft is actually moving with respect to any pickup, the same motion and direction will be duplicated on the CRT.

When using non-contacting pickups with a power supply with (+) polarity, if the vertical pickup is mounted below the shaft and the horizontal pickup on the left, the shaft motion will be read correctly. If they are mounted the other way, the polarity of the input signal to the scope needs to be reversed to read correctly.

When using non-contacting pickups with a power supply with (-) polarity, if the vertical pickup is mounted above the shaft and the horizontal pickup on the right, the shaft motion will be read correctly. If they are mounted the other way, the polarity of the input signal to the scope needs to be reversed.

If this is not carefully done, it will most likely lead to false interpretation of the data. Thus, closely examine the specifications of non-contact pickups installed on machines to be analyzed using orbit pattern analysis.

8.07 SETTING UP THE OSCILLOSCOPE FOR LISSAJOUS ORBITS

Proper calibration of the horizontal and vertical gain controls of the oscilloscope is important to ensure that the shape of the observed pattern accurately reflects the true motion of the shaft, and to make it possible to obtain accurate amplitude data from the scope. While most scopes have separate horizontal and vertical gain controls, using a scope incorporating matched and locked horizontal and vertical gain adjustment can greatly reduce setup and calibration time. In any event, where system sensitivity is known, calibration of the scope gain settings is simply a matter of setting the oscilloscope gain sensitivities to match input sensitivities. For example, for a 200 mv/

mil system, a typical calibration is to adjust the scope to read 200 millivolts per division. In this way, each division on the scope represents 1 mil of displacement. If additional gain is needed, the scope can be set to even submultiples of pickup sensitivity. Thus, for a 200 mv/mil system, setting the gain controls at 100 mv/division would yield 0.5 mil per division or 1.0 mil for 2 divisions. Again, the horizontal and vertical gain controls must be in agreement so that 1 mil of vertical motion produces the same indication on the CRT as 1 mil of horizontal deflection. The scaling factor (i.e., mils/division) should be recorded for future reference. (Ref. 2)

Also to enable the analyst to visualize the motion of the shaft in relation to the position of the non-contacting pickups, he may wish to mark lines on the scope screen to represent the actual horizontal and vertical axis of the bearing. Especially if the display was rotated 45° on the CRT (due to the fact that the pickups are 45° from the horizontal on the machine), it would be desirable to do this, as shown in Figure 2.

8.08 INTERPRETING LISSAJOUS ORBITS WITH OR WITHOUT "BLANK" SPOTS

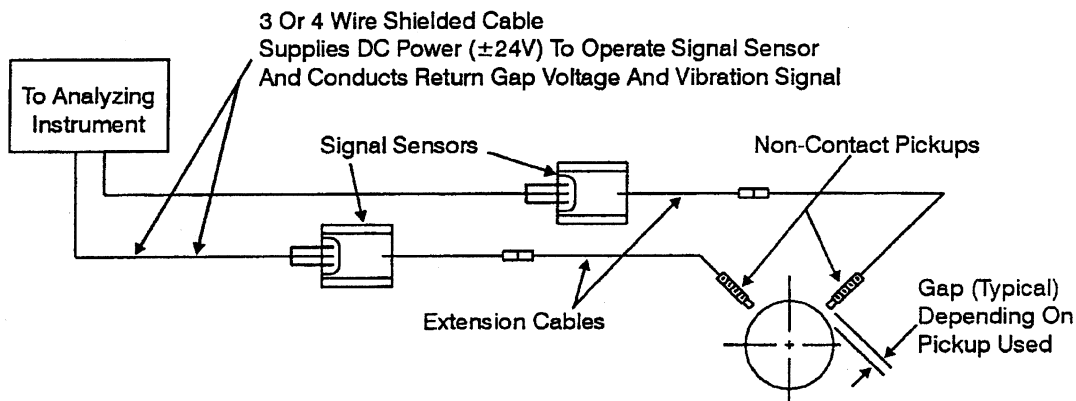
Sometimes Lissajous orbits are shown with "blank spots" superimposed on their orbits. At other times no blank spots or keyphasor marks are shown. The problem is, that if no blank spots are present at all, it becomes extremely difficult to determine whether a harmonic frequency is predominant or whether a subharmonic is being displayed. In other words, if no keyphasor reference is used, it will not be possible to determine whether the loops are harmonically or subharmonically related.

In general, if the pattern from 1X RPM is being featured the orbit is known as the displacement orbit. If a trigger, such as a keyphasor is used, it will trigger a once-per-revolution reference pulse which will be superimposed on the displacement orbit as a "blank spot". If a secondary loop also has a blank spot superimposed on it, the secondary loop represents a subsynchronous component of the rotor's motion. If the blank spot is "locked in-phase" with the blank spot on the displacement orbit, then the subsynchronous component is harmonically related. If the blank spot moves in relation to the blank spot on the displacement orbit, or if the secondary loop moves also, then the secondary loop is not harmonically related; it is nonsynchronous.

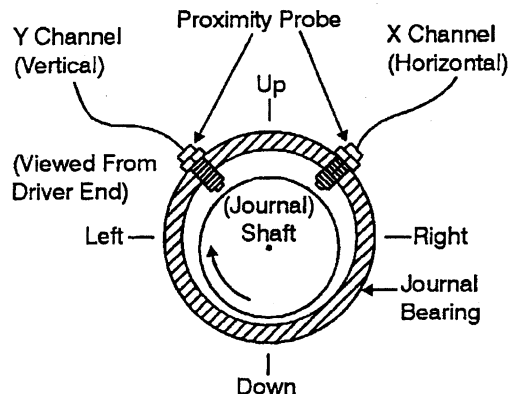
In general, external secondary loops refer to harmonic orders whereas internal secondary loops represent subharmonic components. However, if each of the external secondary loops have a "blank spot", they then represent a particular dominant subharmonic frequency in this case. That is, a "Figure 8" pattern with each orbit having a "blank spot" represents the 1/2X RPM component of the rotor's motion. If there are four orbits with each orbit having a "blank spot", it represents the 1/4X RPM component of the rotor's motion. An excellent example of this is shown in Chapter 5 in Figure 2.

On the other hand, if the orbits represent a particular harmonic multiple, then only the predominant frequency orbit, usually 1X RPM orbit, has the blank spot on it, whereas the other external loops will not have any "blank spots". That is, if the displacement orbit representing the 1X RPM motion has a "blank spot", the external secondary loop represents the 2X RPM component if it remains stationary with respect to the 1X RPM orbit portion. If not, then the external loop is not synchronous with the dominant 1X RPM motion of the shaft. In either case, if the orbit does not have a "blank spot" for higher than 3X RPM components, the orbit becomes complex and beyond the scope of this manual.

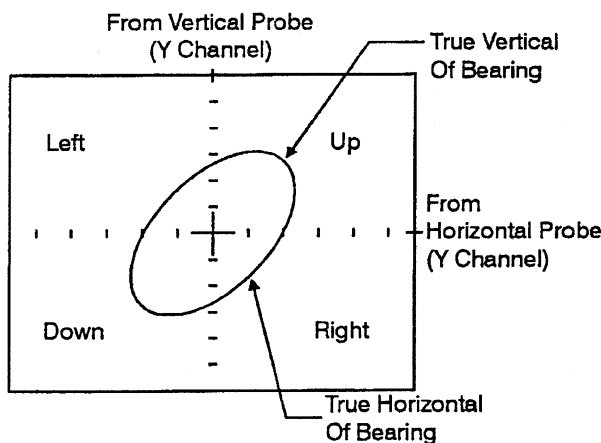
In other cases, the orbit may have multiple "blank spots" superimposed on it rather than internal secondary loops, or external loops, each with a "blank spot". That is, a Lissajous orbit with two "blank spots" means that the 1/2X RPM orbit is being represented; three "blank spots", 1/3X RPM; four "blank spots", 1/4X RPM; and so forth. It is important to state again that there can be an endless variety of orbits, particularly from rubs, and this chapter is intended to be only introductory to this diagnostic tool.



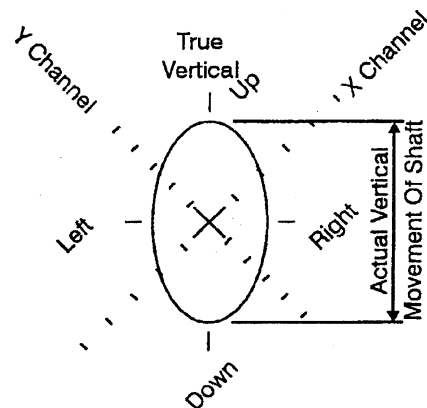
A TYPICAL DUAL PICKUP SETUP USING SIGNAL SENSORS, INSTEAD OF A NON-CONTACT DISPLACEMENT ACCESSORY (SIGNAL CONDITIONER)(Ref. 2)



According to specifications developed by American Petroleum Institute, the position of the probes in the bearing housing should be viewed from the location of the driver. Looking from the driver at the driven, the probe that is on the right hand side should be noted as the horizontal or x-channel probe. The direction of rotation of the shaft is not important in designating which is the "horizontal" and which is the "vertical" probe.



A POSSIBLE LISSAJOUS ORBIT AS DISPLAYED ON AN OSCILLOSCOPE SHOWING THE SCREEN'S VERTICAL AND HORIZONTAL AXIS OR CHANNELS.



MENTALLY ROTATE THE DISPLAY TO DUPLICATE THE SETUP AT THE BEARING. THE MAJOR AND MINOR AXIS CAN BE MEASURED WITH A SCALE OR THE GRID ON THE CRT.

**FIGURE 2
SETUP AND MEASUREMENT OF LISSAJOUS ORBITS (Ref. 3)**

8.1 TYPICAL LISSAJOUS PLOTS FOR COMMON PROBLEMS

The machines to which the analysis of Lissajous plots applies are likely those with journal bearings and non-contact eddy current pickups positioned close to the shaft (usually at a gap of approximately 40 to 60 mils). It is assumed that both pickups have been calibrated and installed properly. If more definition of machine type is required, let the term "machinery" refer to high-speed turbomachinery such as turbines, compressors, and centrifugal pumps. The pickups are installed 90° from each other - one in the horizontal position and the other in the vertical position for simplicity (again; many such probes are oriented at the 10:30 and 1:30 clock positions). The reference pulse giving the once-per-revolution signal is provided by another non-contacting pickup triggering off the key in the shaft. It is assumed that this pickup has also been set up properly. The plots which follow have been modified to simulate the "blank" spot on the orbits which indicates the position on the shaft (via the key).

8.11 UNBALANCE

Ideally, the Lissajous plot for a pure unbalance condition would be a perfect circle. Of course, the diameter of the circle would increase proportional to the increase in operating speed vibration amplitude. There would be one "blank" spot from the reference pulse. Figure 3 is an ideal example of a Lissajous orbit for an unbalance condition.

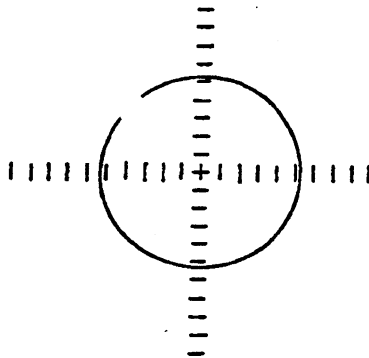


FIGURE 3
EXAMPLE OF A LISSAJOUS ORBIT INDICATING AN
UNBALANCE CONDITION (Ref. 5)

Although it is possible for real-world machines to closely approach this ideal condition, the much more common Lissajous plot indicating unbalance is shown in Figure 4. To have a perfect circle, the horizontal and vertical stiffness of the bearing would have to be identical. More commonly, machines are stiffer in the vertical (normal; perpendicular) direction because they are bolted to the foundation or base in that direction. The effect of gravity is a force also acting in the vertical (downward) direction. Thus, the Lissajous orbit will be somewhat elliptical in these cases due to unequal stiffness or resistance to motion.

How elliptical can the orbit be before it would no longer indicate an unbalance problem? There are no hard-and-fast rules. In general, if the ratio of the major axis to the minor axis is between 3:1 and 5:1, the Lissajous orbit could still be a result of an unbalance condition. If the major to minor axis ratio is higher than 5:1, then the condition leans more towards another problem other than unbalance such as misalignment, resonance, eccentricity or excessive clearance due to bearing wear. Remember, "there is always an exception to the rule".

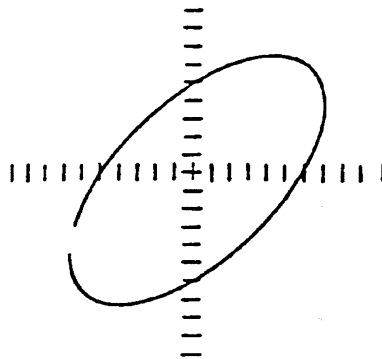


FIGURE 4
AN EXAMPLE OF A MORE LIKELY LISSAJOUS ORBIT INDICATING UNBALANCE (Ref. 5)

8.12 MISALIGNMENT

If there is a shaft misalignment condition, especially across the coupling, the Lissajous plot may resemble an ellipse as shown in Figure 5. In general, the ratio of the major axis to the minor axis is from 3:1 to 8:1 for the elliptical orbit in a misalignment situation, with 5:1 or 6:1 being the norm. However, unlike unbalance, the 1X RPM contribution may be coupled with harmonic orders, 2X, 3X, or higher orders of shaft speed, particularly if mechanical looseness is also involved. Figure 6 is an example of a Lissajous plot from a misalignment situation, possibly coupled with a looseness or shaft "end play" influence. Notice that the ellipse is distorted. It is not a perfect figure eight. The length of the "tail" will indicate the relative amplitude level contributed by the harmonic (distortion) components. If the 1X RPM component is one half or twice the 2X RPM component in amplitude, a "figure 8" orbit as shown in Figure 7 may be formed. In this case also, the pattern and the "tail" will be influenced by the phase and the relative amplitudes of any other harmonic components which may be present.

In cases of excessive misalignment, other shapes are possible. There may be a truncated orbit, shown in Figure 8(a); a banana-shaped orbit, shown in Figure 8(b); and others, as shown in Figure 8(c). These more complex shape examples result from the distortion of the 1X RPM displacement component with those from the higher harmonic orders.

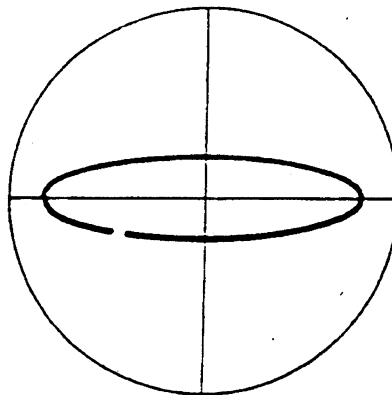


FIGURE 5
A LISSAJOUS ORBIT EXAMPLE FOR A MISALIGNMENT PROBLEM (Ref. 2)

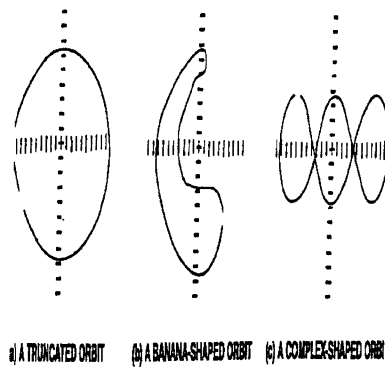


FIGURE 6
A "FIGURE 8" LISSAJOUS ORBIT EXAMPLE FOR A COMBINED MISALIGNMENT AND LOOSENESS PROBLEM (Ref. 5)

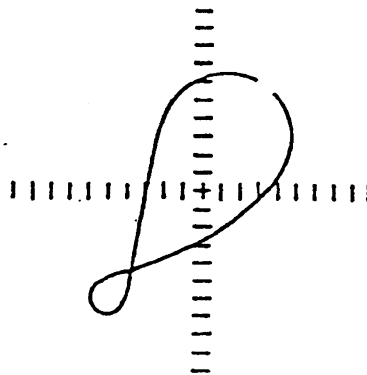


FIGURE 7
AN EXAMPLE "FIGURE 8" LISSAJOUS ORBIT FOR A MISALIGNMENT PROBLEM WHERE 1X RPM & 2X RPM HAVE A 2:1 RATIO (Ref. 5)

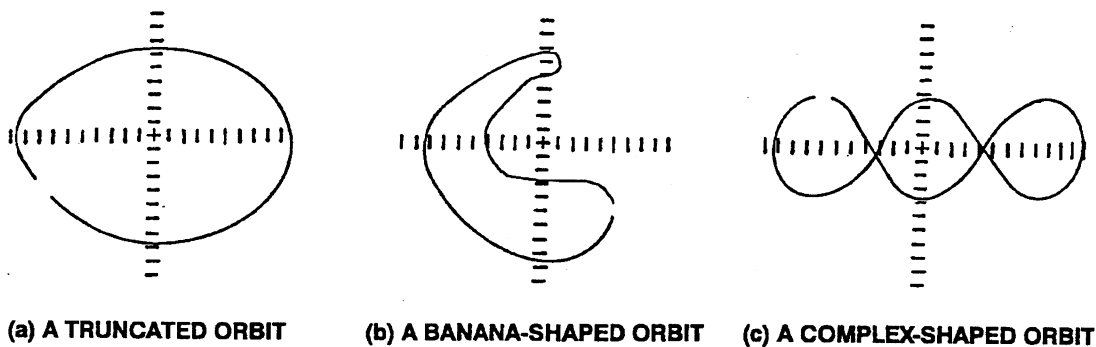


FIGURE 8
VARIOUS EXAMPLES OF LISSAJOUS ORBITS FROM CASES OF EXCESSIVE MISALIGNMENT

8.13 HOW CAN IT BE DETERMINED WHETHER THE LISSAJOUS ORBIT IS CAUSED BY UNBALANCE, MISALIGNMENT OR RESONANCE?

There is no "surefire" way to differentiate between these problems with orbit analysis alone. A key to the solution would be the ability to change the shaft speed higher or lower. Another key would be to watch the change in position of the reference "blank" spot as the speed is changed. For example, to determine if the orbit is caused by unbalance or resonance, (particularly if the shape is elliptical), watch the position change (orientation) of the "blank" spot. If it is assumed that the shaft is running below its first critical speed and the shaft RPM is increased, the "blank" spot on the orbit will shift as shown in Figure 9. At resonance the "blank" spot will shift approximately 90° away from where it was before the shaft speed was changed. If the speed continues to increase past the first critical, the "blank" spot will shift approximately 180° away from its original location as shown in Figure 9d and 9e. However, it was only assumed that the shaft was operating below its first critical. If the shaft speed is located between the first and second critical, a similar position change of 180° in the "blank" spot could be observed as the shift decreases in speed to below its first critical. Then, it could be concluded in both cases, that the elliptical shape was due to the shaft speed being in close proximity to its critical speed. It would not be due primarily to an unbalance condition. If the "blank" spot does not move as the speed is increased, then it could be concluded that the problem is from unbalance or possibly from misalignment. Also, the shape of the ellipse will stay basically the same with unbalance but the major and minor axis will increase proportional to the increase in vibration amplitude.

To determine whether the elliptical orbit is from misalignment or resonance, observe the change in the shape of the orbit as the speed is varied. If the shaft is running near its critical speed, then the shape of the Lissajous plot will change noticeably if the shaft speed is increased or decreased as shown in Figure 9c. If the ratio of the major axis to the minor axis is 8:1 to 10:1, or more, this strongly suggests that the rotor is operating near its critical speed or even possibly near a structural resonance. Once again, watch for the change in shaft speed to affect the shape of the Lissajous orbit. (Ref. 2)

These suggestions are only meant to be guidelines, not hard-and-fast rules. Machinery dynamic behavior is very complex; often, solutions are not this simplistic. For instance, in many cases it is not an easy process to increase or decrease the shaft speed. If this is not expedient or practical, a judgement call is often made based on the available diagnostic information on hand.

8.14 ROTOR RUBS

If a machine is subjected to a rotor rub, it has no internal corrective responses with which to eliminate the effects of the rub. A rub is generally caused when the rotor contacts a stationary component, such as the bearing surface, a seal, and in the case of a motor, the stator. There are many causes of rubs with varied effects. Some common causes are: excessive unbalance, misalignment, soft foot, excessive clearance in the bearing, thermal bow, and others. Rotor rubs are basically one of two types: partial rubs and full rubs.

8.141 Very Mild Rotor Rubs

A very mild rotor rub is one in which the shaft merely "touches" the stationary component once per shaft revolution. The result is a slight distortion of a circular and elliptical orbit as shown in Figure 10. Along with this slight distortion, there will be a back-and-forth movement of the "blank" spot in the Lissajous orbit. This movement of the "blank" spot representing a keyphasor oscillation is called keyphasor "jitter", particularly when the position of the "blank" spot changes by more than 10°. However, mild rubs often deteriorate into full rubs over time. (Ref. 4)

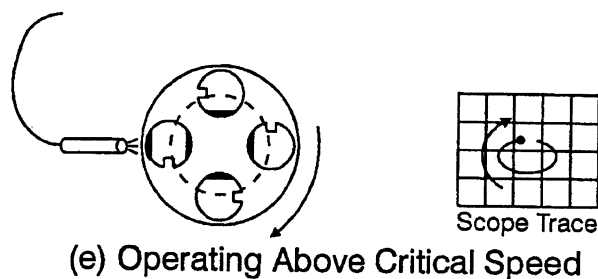
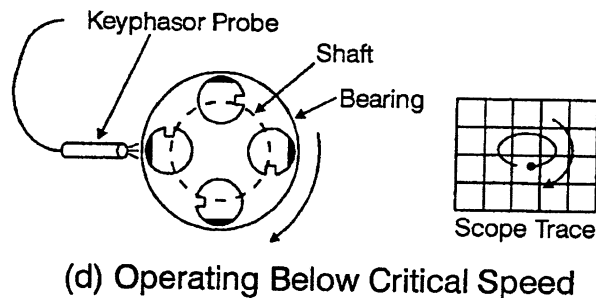
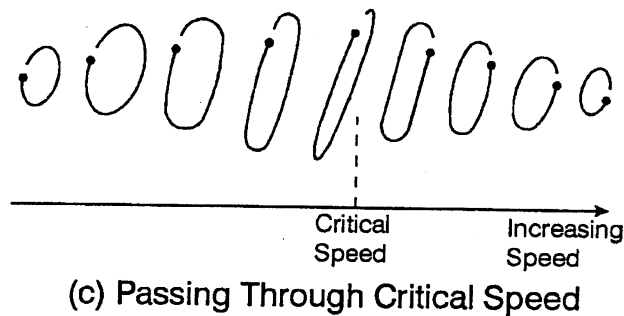
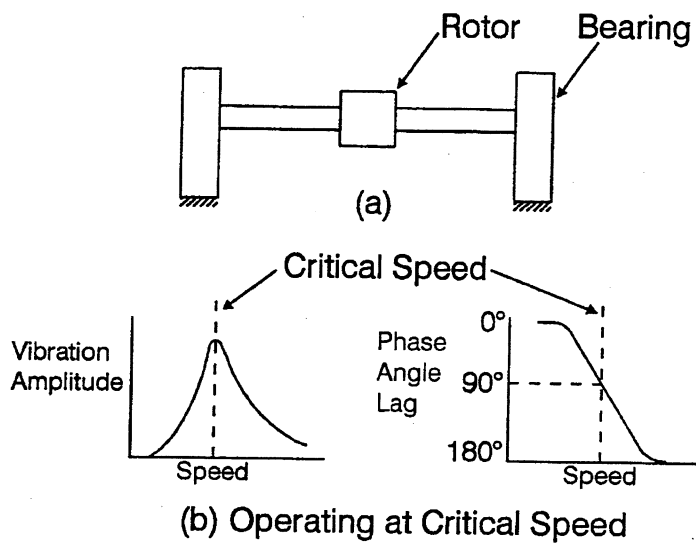


FIGURE 9
OPERATIONAL CHARACTERISTICS OF A SIMPLE ROTOR SYSTEM (d) BELOW AND
(e) ABOVE THE FIRST CRITICAL SHOWING THE EFFECT ON LISSAJOUS ORBITS

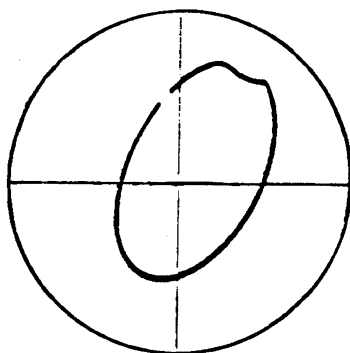


FIGURE 10
AN EXAMPLE OF A LISSAJOUS ORBIT FROM A VERY MILD ROTOR RUB
 (Ref. 2)

8.142 Heavy or Full Rubs

When the rub becomes more severe, the Lissajous orbit tends to become increasingly more complex and erratic. Component resonances can be excited, increasing orders of harmonic frequencies can appear, and other nonsynchronous frequencies interact to create an endless number of possible effects. Figure 11 is a representative example of various patterns of Lissajous orbits for heavy or full rubs.

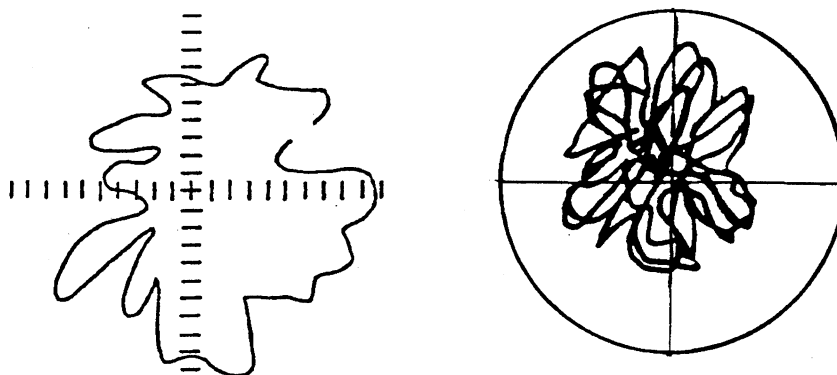


FIGURE 11
EXAMPLES OF LISSAJOUS ORBITS FOR HEAVY OR FULL ROTOR RUBS
 (Ref. 2, 4)

By observing the oscillations of the "blank" spot, an insight into the severity of the rub can be obtained. As the rub degenerates into a worse condition, the movement of the "blank" spot will become more random and erratic. It is a reflection of what is occurring to the shaft (via the key) as the rub takes place. Initially, with a slight rub, there is a slight resistance to the direction of shaft rotation. As the amplitude level increases because the situation is getting worse, there is a tangential force which acts on the shaft where it contacts the bearing or seal in a direction opposite to shaft rotation. At the time the shaft actually contacts the stationary surface increases, a reverse precession will be observed in the Lissajous orbit. When the tangential force becomes large (as in the case of a full rub), the shaft will orbit in a direction opposite to the direction in which it is driven to rotate. In most cases, a shaft which is experiencing a full rub condition will lead to a catastrophic failure of the machine. Figure 12 is an example orbit of this reverse precession during a full rub. The presence of 4 "blank" spots indicates that the frequency of the orbit is $1/4 \times \text{RPM}$.



FIGURE 12
AN EXAMPLE LISSAJOUS ORBIT OF THE REVERSE PRECESSION IN A
FULL ROTOR RUB (Ref. 4)

8.143 "Hit and Bounce" Rubs

Between the extremes of a mild rub and a full rub is a family of partial rubs, known as the "hit-and-bounce" variety. A condition such as misalignment can cause a shaft to lightly touch the bearing surface which suppresses the rotor's tendency to move outward radially. However, as the condition degenerates, more happens to the orbit other than a mere flattening at one point-per-shaft revolution. The shaft will hit and bounce off. This can result in a reverse whirl. Most often this occurs at exactly 1/2X RPM. Reference 2 provides the following helpful information: "two full revolutions of the shaft have to be set up in the instrument to be able to display this Lissajous orbit".¹ This will be shown in the orbit as a secondary *internal* loop. In another setup, two stationary "blank" spots may be present on a single orbit. This will indicate that the frequency is 1/2 the shaft RPM (Ref.2). As the condition degenerates, more and more hits can occur and they will be reflected in the Lissajous orbits, as shown in Figure 13. Notice that as the hits become more frequent, the internal loops become tighter and tighter.

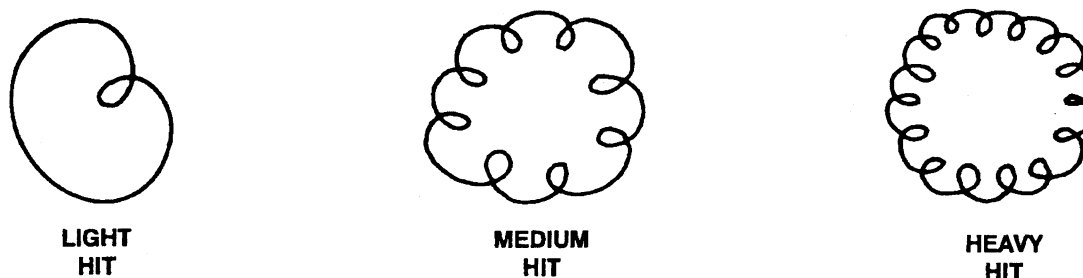


FIGURE 13
EXAMPLE LISSAJOUS PLOTS OF VARIOUS "HIT-AND-BOUNCE"
ROTOR RUBS (Ref. 4)

8.144 Conclusions From Lissajous Orbits in Rotor Rub Diagnostics (Ref. 3)

For partial rubs:

1. They create orbits with internal loops.
2. They cause the "blank" spot to jitter.

For full rubs:

1. The orbit rotation is *opposite* the direction of shaft rotation (reverse precession process).
2. Even though the shaft may slow down due to the friction in the rub, the frequency the reverse rub orbit is virtually constant.

8.15 OIL WHIRL

Oil whirl usually occurs in machines with journal bearings and a pressurized lubrication system. Generally, the shafts are long and have relatively high RPM's (operate above their first critical). The bearings are lightly loaded with low damping. Unbalance or misalignment usually provides the excitation force that supplies the impetus behind oil whirl.

The shaft has a tendency to ride on wedge of oil which has a lifting effect on the shaft. This "small destabilizing force [combines with the excitation force to] cause the shaft and lubricant to circulate strongly in the direction of the shaft rotation",¹ which causes the shaft to whirl as it rotates. The oil whirl frequency will be between 40% and 48% of the shaft speed.

The Lissajous orbit for oil whirl condition will show a secondary internal loop that slowly rotates about the axis of the displacement orbit. It rotates at a frequency independent of the displacement orbit because it is not synchronous with rotation. Figure 14 is an example of a Lissajous orbit from a shaft subjected to an oil whirl condition. (Ref. 2)

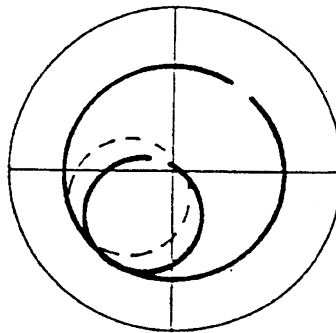


FIGURE 14
AN EXAMPLE LISSAJOUS ORBIT FROM AN OIL WHIRL CONDITION (Ref. 2)
(Internal Loop Slowly Rotates About the Axis of the Orbit)

8.16 MECHANICAL LOOSENESS

The Lissajous orbit behavior due to mechanical looseness is similar to that for a rub. The most common cause of mechanical looseness is too much clearance in the bearing. The outstanding effect of mechanical looseness is the presence of integer fraction subharmonics of 1X RPM, (usually 1/2X or 1/3X), as well as the presence of harmonic orders of 1X RPM of the shaft. The subsynchronous frequencies will produce secondary internal loops similar to the Lissajous orbits for rotor rubs. Of course, in more complex situations, the orbits will be quite erratic, as shown in Figure 15.

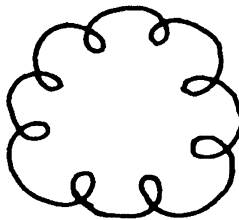


FIGURE 15
AN EXAMPLE LISSAJOUS ORBIT SHOWING COMPLEX INTERNAL LOOPS
DUE TO MECHANICAL LOOSENESS

8.17 HOW CAN IT BE DETERMINED WHETHER THE LISSAJOUS ORBIT IS FROM MECHANICAL LOOSENESS, ROTOR RUB OR OIL WHIRL?

To determine whether the problem is from mechanical looseness or a rotor rub, the key to the solution is in determining the direction that the displacement orbit moves in relation to the shaft rotation. If the problem is mechanical looseness, the displacement orbits and secondary loops will be locked "in-phase" regardless of the direction the shaft is rotating. If the problem is rotor rub, the displacement orbit will have a rotating tendency (via the "blank" spot) in a direction opposite to shaft rotation, especially in full rub situations.

To determine whether the problem is from mechanical looseness or from oil whirl, consider the secondary loop. If the problem is oil whirl, the internal loop will slowly rotate in the same direction as shaft rotation. If the problem is mechanical looseness, the secondary loops will be stationary since they will be synchronous with the 1X RPM displacement orbit. This "in-phase" feature will also apply to rotor rub.

8.2 APPLICATIONS OF LISSAJOUS ORBITS NOT COVERED

There are more applications and diagnostic features of Lissajous orbits which are beyond the scope of this chapter. It was intended to be an introduction to the use of Lissajous orbits for machinery diagnostics. Lissajous orbits are also used to measure integral ratios between sine waves of different frequencies. The examples and Lissajous orbits used in the chapter were basic. In the real world, the non-sinusoidal effects from gear imperfections, bearing faults, electrical vibrations and so forth, will cause "hash" on the waveforms. One can imagine the effects on the Lissajous orbits. In these cases, sophisticated instrumentation and persons with great expertise may be required to separate the real problems from all the background "noise", as well as to evaluate the severity of problems found.

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